BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 289 · Hashing and applications

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the encryption key in symmetric encryption?

Q2. What is the role of a public key?

Q3. Why encrypt a message in transit?

Q4. Compute 10110110_2 AND 00001111_2, then hexadecimal.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

hash — Fixed-size digest of arbitrary input.

collision — Two inputs producing the same digest.

salt — Random value mixed into password hashing.

KDF — Key/password derivation function.

integrity — Confidence that data has not changed.

New learning

Understand — A cryptographic hash transforms an arbitrary input into a fixed-size digest. It is designed to make reversing the digest infeasible, not impossible under all circumstances.

Trace — A message digest supports integrity checking when the reference hash is trusted. Hash collisions can exist because outputs have finite length.

Evaluate — Passwords should be processed using a salted, purpose-built slow password-hashing scheme, not simply a fast general-purpose hash and never reversible encryption.

Worked example

A downloaded file's trusted SHA-256 checksum differs from the computed checksum. What can you conclude?

1. Predict first — The file contents or expected reference differ: the integrity check fails.

2. Show method — The mismatch does not by itself reveal whether alteration was accidental or malicious.

We Do · Practise with guidance

Two users choose the same password. How can salts improve their stored password verifiers?

```
verifierA = KDF(saltA, password)
verifierB = KDF(saltB, password)
```

Predict first

Different random salts make identical passwords produce different stored verifiers.

Show method

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A costly password KDF slows guessing; salt is generally stored beside the verifier and need not be secret.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define cryptographic hashing and distinguish it from reversible encryption. [6]

Question 2

Explain how a trusted checksum helps verify a software download and give one limitation. [6]

Question 3

Explain salted password hashing and why storing raw passwords or just fast unsalted hashes is weaker. [6]

Exit Ticket · Answer before checking

If a password is hashed, the server simply decrypts that hash during login.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A shared secret used to encrypt and decrypt.

Q2. Can be distributed for encryption or verification according to the scheme.

Q3. To protect its content from unintended readers.

Q4. $00000110_2 = 06_{16}$; high four bits masked away.

You Do pages · Independent questions

Q1. Hash maps data to fixed-length digest [2], designed one-way [1]; encryption designed to recover plaintext with key [2]; hashes are not decrypted [1].

Q2. Compute digest of downloaded bytes [2]; compare with trusted expected digest [2]; match is strong integrity evidence [1]; compromised reference cannot establish authenticity [1].

Q3. Unique random salt per account [2], purpose-built slow KDF [2], prevents direct same-password verifier equality/precomputed attacks [1], raw or weak unsalted hashes expose credentials [1].

Exit Ticket · Model answer

False. It computes a candidate verifier and compares it.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding